

Lazy Loading

Ship less JavaScript on first load — a route's code loads only when it's actually visited, instead of shipping every component inside the **main bundle** up front.

1 · THE PROBLEM

Every route ships eagerly, whether it's visited or not

With a normal eager import, Angular bundles a route's component into the main JavaScript file the browser has to download and parse before the app can even start — even for routes the user may never open.

2 · ONE-LINE CHANGE, REAL PAYOFF

Eager imports vs. `loadComponent`

✗ Before — every component is in the main bundle

```
import { Products } from
  './exercises/product-card/product-card';
import { Checkout } from
  './exercises/checkout/checkout';

export const routes: Routes = [
  { path: 'products', component: Products },
  { path: 'checkout', component: Checkout },
];
```

✓ After — code loads only when the route is visited

```
export const routes: Routes = [
  { path: 'products', loadComponent: () =>
    import('./exercises/product-
      card/product-card')
      .then(m => m.Products) },
  { path: 'checkout', loadComponent: () =>
    import('./exercises/checkout/checkout')
      .then(m => m.Checkout) },
];
```

3 · WHY IT MATTERS

Smaller initial bundle, faster first paint

1

Smaller main bundle

The browser downloads and parses less JavaScript before the app becomes interactive.

2

Loaded on demand

A route's chunk is fetched only the moment the user actually navigates to it.

3

One-line change

`component: X` becomes `loadComponent: () => import(...)` — no restructuring needed.